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Development of a mobile application with embedded LLM for hiking and trails

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1 Introduction

Hikers face a general problem while planning their activities: finding one or more trails suitable for their interests [1]. This is the primary motivation for the development of this project.

Using a Large Language Model (LLM) [2], the application will provide, in real time, many trail suggestions for the user based on their request (through a chatbot interface or defined parameters). Users will be able to chat with a chatbot for assistance within the application. The system may also support importing trails from other applications (optional). Based on previous user trails, there can be suggestions for new similar routes or previous ones, while also providing a history for later reference.

Hikers can also share trails with friends, promoting a collaboration network for other hikers to explore new spaces or share statistics. This friend network also allows hikers to set a group trail, promoting social interaction [3].

2 Review

The application must have the following requirements:

2.1 Functional Requirements

- An interactive map using a map SDK with the purpose of users being able to see their location and the current trail and/or trails around them [4];
- Read the location of the users through the mobile device being used;
- Allow users to create, view and initialize trails alone and/or in group, keeping track of user's location and having available different statistics. Each trail must have different checkpoints for the user to pass by to finish the trail;
- Provide a chatbot ("Gryllie", inspired by famous adventurer "Bear Grylls"), using an LLM as the main component, to help users in the application's context. This can be via written text or speech-to-text mechanism;
- Allow users to create and authenticate themselves via OAuth/OpenID [5] [6] and/or user name and password;
- Allow users to test new trails set by the community and validate them by enough reviews or a certain amount of time;
- Trails can have different classifications regarding difficulty (such as "Beginner", "Intermediate", "Advanced") and type (such as "Public", "Private", "In Test", etc.);
- Optional:
  - Friendship system, that consists of a user having friends within the application to share trails, start group trails, etc.;
  - Suggesting trails to users based on previous routes taken;
  - Evaluate the weather conditions of the trail's area to advise the user, using weather API;
  - During trail execution, show medium speed, actual speed, kilometers, etc.

2.2 Non-Functional Requirements

- A scalable web API, using load balancing mechanism;
- Allow future extension to iOS (impossible to test at the moment);
- Unit and integration tests to not have inconsistency within the application;
- High availability to ensure that users can access any information at any time;
- High performance to ensure users do not have to wait too much time for their requests.

2.3 Technologies and tools

For client-side business logic, the following technologies will be used:

- **Compose Multi-platform** along with **Kotlin Multi-platform (KMP)** for the non-web UI/UX and business logic, namely Android and possibly iOS;
- **React** along with **HTML**, **CSS** and **TypeScript** for the web UI/UX (following the *Single Page Application* pattern) or **Kotlin Compose for WASM** (depending on having web support).

Since the application will be accessed across the world through the different types of clients specified above, a web API must be developed to accommodate that need. In order to do so, the following technologies will be used:

- **Spring Boot** along with **Kotlin** for the web API's business logic;
- **PostgreSQL** for database management;
- **JDBI / Exposed (Kotlin)** for the intercommunication between database and web API;
- **nginx** for web proxying and load balancing;
- **Docker** for system isolation and load balancing.

During the development phase, the following frameworks will be used:

- **Vite** for web front-end testing and bundling (depending on having web support);
- **JUnit** for web API & multi-platform front-end testing;
- **Gradle** for primarily building and package management of the web API and non-web front-end environments.

2.4 Problems and Risks

One of the possible risks induced by this project is the usage of LLMs; since they are usually prone to hallucination, they can provide wrong answers to the end user.

Other risks to have in mind are the inexperience of the group members, both in building a scalable API and in the working process behind LLMs; however, we are confident on the application of the knowledge provided by the Data Science Fundamentals course and in the failure we had in building an scalable API in the Web Application Development course to combat said risks.

3 Calendar

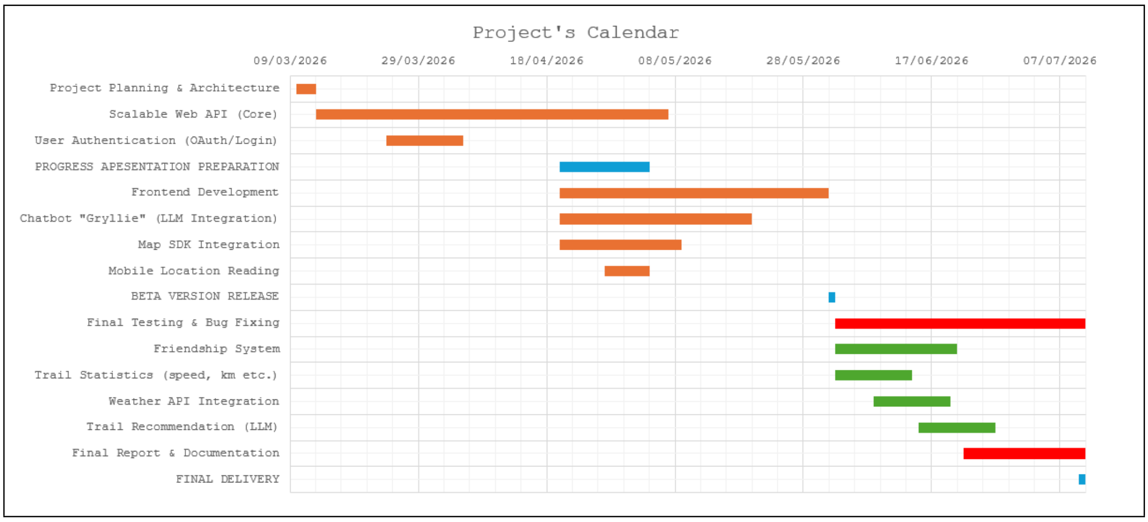

Figure 1: Project Calendar for every work week

References

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